

(PROJECT NAME)

Ten Minutes, No Jargon

The same pitch, written for judges who are not engineers. Every number is one we measured ourselves. Every technical term has been swapped for an everyday one — with the technical version kept on the last page, in case someone asks.

TOTAL RUNTIME

10:00

PACE

~140 wpm

SLIDES

8

JARGON WORDS

0

This version assumes the panel includes people who fund things, administer districts, or judge on impact — not people who write solvers. So it explains **what the thing does for a person in a village**, and saves the machinery for the questions.

Two rules while speaking. **Never use a word you would not use with your grandmother.** And **when you say a number, immediately say what it means** — "forty-six minutes, so you finish before the water gets there" beats "forty-six minutes" every time.

READING THE CUES

PAUSE — stop for a full second and look up. **Bold** — land this word. **Green figures** — a measured number, say it precisely. Blue boxes are delivery notes. Amber boxes are what to say if someone pushes back.

Good morning. We are Team Bit Loss, and we are working on Problem Statement 26161 — predicting the flood that follows when a dam breaks. Our project is called **(PROJECT NAME)**.

Here is the situation. When a dam gives way — or when a landslide blocks a river and that natural blockage bursts — the villages downstream have between twenty minutes and two hours before the water arrives.

PAUSE

Today, preparing the flood prediction takes longer than the flood does. Not because computers are slow. Because **a trained specialist has to set the whole thing up by hand**, and that takes days or weeks. **That gap is the problem.** Let me show you how we close it.

Good flood software already exists. Engineers use it and it works well. The trouble is that it was built for **studies**, not for **emergencies**. Five reasons why.

One — a person has to prepare it, and that takes weeks. Before the computer does anything at all, an expert has to collect the map of the land, mark out the river, and set up

the whole problem by hand. Most districts do not have that expert sitting free and waiting.

Two — it needs the internet, and disasters take the internet down. Most modern tools live in the cloud. The network is usually the first thing to fail in a flood, and it fails exactly when you need the answer.

Three — nobody is watching the mountains. When a landslide blocks a river high in the Himalayas, there is no sensor there and nobody living nearby to report it. A lake quietly builds up behind the rubble for days — and then it lets go. That is roughly how the Chamoli disaster began in 2021.

Four — the map is out of date the moment it matters. A landslide physically changes the shape of the valley. But the software is still using an elevation map made years earlier. So it gives you an answer that is wrong — and it sounds **completely sure of itself**, which is the dangerous part.

Five — you get one number, with no honesty about the uncertainty. One figure for how much water. No range. No worst case. And a district officer has to decide whether to evacuate ten thousand people based on a single number that nobody can put a margin of error on.

PAUSE

So: slow to set up, fragile, blind to landslides, working off old maps, and overconfident. That is what we set out to fix.

DO NOT SAY THIS

Avoid "existing tools take two to six hours to run." If a judge asks how long ours takes, our honest answer for a detailed run is **five hours** — and then the criticism lands on us instead. The gap we genuinely close is the **weeks of human setup**, so make that the point.

SLIDE 3 / 4

What We Built, and Why It Is Different

2:45 – 5:15 · 2.5 min

This is (PROJECT NAME). Let me walk you through what actually happens, step by step.

You type in four things.

Where the dam is, how tall it is, how much water it holds, and what kind of failure you want to test. That is the entire input. Four boxes on a screen.

PAUSE

Compare that with weeks of an expert's time.

It goes and collects its own data.

The shape of the land, where people live, satellite pictures, what the ground is covered with — forest, farmland, city. All of it comes from free public satellite data that anyone can download. No logins, no fees, nobody sent into the field with a survey instrument. **The job that used to take an expert a week, the software now does by itself in minutes.**

It doesn't guess how the dam breaks. It tests every reasonable guess.

Nobody knows in advance exactly how a dam will fail — how wide the hole will be, how fast it opens. So instead of picking one guess and hoping, we take the published research on how dams have actually failed in the past, and we run the flood **several times over, changing the guess each time**. That way we are not betting on one assumption. We are covering the whole range that history says is possible.

Then it simulates the water — using two different methods.

Right at the dam, the water is violent and chaotic, so we use a detailed method that follows the water almost droplet by droplet. Further downstream, the flood spreads out into a wide sheet moving down the valley, and there a much faster method gets it right. **Using the heavy method everywhere would take days. Using the light method everywhere would get the dam itself wrong. So we use each one where it belongs.**

And it tells you how confident it is.

Because we ran the flood many times over, we can give you three answers instead of one: **the best case, the most likely case, and the worst case**. An officer deciding on an evacuation can plan against the worst case rather than a single number with nothing behind it.

And we checked our answers against someone else's.

There is a classic dam-break problem where mathematicians worked out the exact correct answer on paper, long ago. We put that problem through our software, and through the professional software that engineers use worldwide — the same problem, both programs, both marked against the known right answer.

PAUSE

We scored slightly closer to the correct answer than the professional package did. This is the part I would like the judges to remember: we are not asking you to take our word for

it. We put our work next to the industry standard and had both of them marked.

Finally, it produces something a real responder can use.

Not equations — maps. How deep the water gets. How fast it moves. And most importantly, **how many minutes until it reaches your village**. These come out in the ordinary map file formats that district offices already open every day, including the format that opens in Google Earth. And the whole thing works with the internet switched off. On the right you can see it running. **This is built and working, not a proposal.**

IF AN ENGINEER ON THE PANEL DIGS IN

Two things to say plainly rather than dress up. The two methods pass information in **one direction only**, not both ways — that was a deliberate scope decision, and saying so is stronger than claiming otherwise. And the checking so far is against the textbook problem, **not** against a photographed real flood. Those real-world comparisons are set up and are our next step.

SLIDE 5

How It Is Built

5:15 – 6:30 · 1.25 min

Very briefly, the engineering — and I will keep this short, because the point is not complicated.

Every single piece of software we used is free and open-source. There is no licence to buy, no annual fee, no vendor to negotiate with. A state disaster authority could install this without a procurement process at all — and for a government system, that is often the difference between a good idea and a deployed one.

It also **does not need special hardware**. No supercomputer, no expensive graphics card. It runs on an ordinary office computer, which is what a district office actually has.

And the mathematics underneath is not ours. It is the standard, published, peer-reviewed maths that this field has used for decades. **We did not invent new physics. We took physics that is already trusted and made it fast and automatic.**

IF YOU ARE RUNNING OVER TIME

Cut the third paragraph. Keep "free and open-source, no licence to buy" and "runs on an ordinary computer" — those two lines are what a government evaluator actually writes down.

Four honest risks, and what we did about each.

Risk one — the detailed simulation is expensive to compute. Answer: we only use the expensive method in the small violent area right at the dam, and the fast method for the long stretch downstream. That is what turns days into under an hour.

Risk two — power and network fail in a disaster. Answer: we download and store everything for your district in advance. After that, it runs with the internet completely dead.

Risk three — the map of the land goes out of date after a landslide. Answer: we update the map ourselves. We take the satellite evidence of the new blockage and write it into the elevation map, so the simulation runs on **today's valley, not last year's**.

Risk four — testing random possibilities can still miss the worst one. Answer: we make sure the combinations where several things go wrong at once stay inside the set we test, instead of being quietly skipped.

PAUSE

None of these needed a research breakthrough. They were engineering decisions, and they are already built into the working prototype.

BE UPFRONT ABOUT THIS ONE

Our updated map is **not a new survey of the valley** — it is the public elevation map with the landslide barrier written into it, and every file we produce says so on its face. Volunteering that limitation makes you look careful, not weak.

Now the part that matters.

The headline is that the waiting-for-an-expert problem disappears. Today, before any prediction can start, a specialist works for days or weeks preparing the data. (PROJECT NAME) does that preparation itself, without a person, in minutes.

PAUSE

And then the computer runs. These are our own measured times, on an ordinary sixteen-core office machine with no special hardware:

46 min

ENOUGH TO ACT ON

A focused area around the dam, simulating 30 hours of flood, six different failure scenarios.

64 min

WIDE FIRST LOOK

A much larger region at coarser detail — the quick overview of where the water goes.

5 hours

FULL DETAIL

The same large region at fine detail. Slower, and we say so — we always publish the settings with the time.

So here is the honest claim, and I want to be precise about it. **The human preparation goes from weeks to minutes, and the computer finishes in under an hour for a run you can act on.** Weeks of setup plus a run means you describe the flood after it has happened. Automatic setup plus forty-six minutes means you describe it **while there is still time to move people.**

IF ASKED WHY THE FIVE-HOUR ONE IS SLOWER

Because it covers an area about nine times larger, not because it is nine times more careful. Always give the settings along with the time — that is what lets someone check us, and offering it unprompted is what makes the number believable.

Four things this changes.

Lives saved — a warning that arrives earlier is a village that gets out in time. **Dangers found that nobody was watching** — we can spot a landslide blocking a river in a valley with no people and no sensors in it. **Less damage** — knowing where the water goes lets authorities move equipment and supplies before it arrives, not after. And **better preparation** — a district can test its dam safety plan on an ordinary Tuesday afternoon, instead of finding out during the emergency.

And where we honestly stand today.

Working now: the simulation is built and tested, it handles both a dam breaking and a landslide blocking a river, it exports the map files a district office can open, and it passes **600** automated tests every time we change the code. **Being worked on:** comparing our results against two real historical floods, live satellite monitoring, and estimating rupee damage — where we currently label our cost figures as unverified rather than pretending they are settled. **Next:** making it work for any river in the country, and producing the emergency response map packs in the exact format the Central Water Commission

already uses — so it fits the government's existing process instead of asking anyone to change how they work.

SLIDE 8

Where This Comes From, and Close

9:30 – 10:00 · 30 sec

None of this is invented in a vacuum. It follows the Central Water Commission's own guidelines for dam flood mapping, the published scientific studies of the Chamoli disaster, and decades of peer-reviewed research on how dams fail and how floods move.

All our data is free and public, and we deliberately avoided any dataset with a restrictive licence — so a government agency can legally share and publish whatever this produces. And every number on this deck can be reproduced: the commands are on the slide.

PAUSE

To close. Existing tools tell you what the flood **was**. (PROJECT NAME) tells you what the flood **will be** — with an honest range, without needing a specialist, on an ordinary computer, with the power and the network down. Thank you.

CARD

Timing card — keep visible

10:00 total

SLIDE	TOPIC	START	LENGTH
1-2	Title	0:00	0:45
3	The problem	0:45	2:00
3 / 4	What we built	2:45	2:30
5	How it is built	5:15	1:15
6	Risks and answers	6:30	1:15
7	What changes on the ground	7:45	1:45
8	Sources and close	9:30	0:30

Three things they should remember

POINT	SAY IT LIKE THIS
Four boxes	"You type in four things. Everything else, the software collects by itself."

POINT	SAY IT LIKE THIS
Weeks → minutes	"The part that needed a specialist for weeks now happens automatically in minutes."
Three answers	"You get a best case, a likely case and a worst case — not one number pretending to be certain."

BACK POCKET

The technical version, if someone asks

Q&A only

Everything above was said in plain words on purpose. If a technical judge asks for the real name of something, here is the swap — say the technical term once, then go straight back to plain language.

WHAT YOU SAID	WHAT IT IS CALLED
"follows the water droplet by droplet"	3D Smoothed Particle Hydrodynamics (SPH), near the breach
"a faster method for the sheet of water downstream"	2D shallow-water solver, HLLC scheme with Audusse hydrostatic reconstruction
"we run it several times, changing the guess"	Monte Carlo ensemble over published breach regressions (Froehlich, MacDonald, Xu-Zhang) with Wahl uncertainty bands
"best, likely and worst case"	5th, 50th and 95th percentiles across ensemble members
"the shape of the land"	Copernicus GLO-30 digital elevation model
"satellites that see through cloud and at night"	Sentinel-1 radar, change detection on GRD imagery
"the map files a district office already opens"	GeoTIFF, Shapefile and KML
"the professional software engineers use"	Delft3D FM, the real Deltares kernel
"a problem where the exact answer is known"	The Ritter dry-bed dam-break solution (1892)
"we scored slightly closer"	0.0317 m error for us, 0.0349 m for Delft3D FM, both against the exact solution

THE THREE THINGS NEVER TO CLAIM

Do not say the two solvers are fully coupled — information flows one way only. **Do not** say we have been validated against real satellite flood photographs — that comparison has not been run yet. **Do not** quote a speed comparison against other software — we have not benchmarked one, and an invented number is the fastest way to lose a technical judge.

(PROJECT NAME) · SIH 2026 · PS 26161 · Team Bit Loss — plain language edition. Every figure here traces to a run log or a test in our repository.